From Transactions to Buildings

Leakage-Aware External Data Enrichment for
Paris Apartment Valuation

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Abstract

This study evaluates whether authoritative building and neighborhood data can improve an automated valuation model (AVM) previously limited to French Demandes de valeurs foncières (DVF) transactions and lagged comparable sales. The fixed cohort contains 143,009 ordinary Paris apartment sales from 2021–2025. Phase 3 links each transaction to Base Adresse Nationale (BAN) address identity; Base de Données Nationale des Bâtiments (BDNB) building attributes and dated energy-performance diagnostics (DPE); rail stations; schools, parks and nearby services; and strategic transport-noise maps. Exact BAN matches cover 99.49% of sales, 99.997% receive a BDNB building identity, and 76.85% receive a DPE that existed by the transaction date. No future-DPE join is observed. Configuration selection uses 2024 only; selected models are refitted on 2021–2024 and evaluated on 28,330 sales from 2025. The selected CatBoost model obtains EUR 93,136 mean absolute error (MAE), EUR 192,806 root mean squared error, $R^2 = 0.8679$, and 11.70% median absolute percentage error. Relative to the frozen Phase 2 comparable-sale correction (EUR 96,428 MAE), the MAE reduction is EUR 3,293, or 3.41%, with a paired 95% bootstrap interval of EUR 2,735–3,846. Dated DPE, building and neighborhood ablations each improve MAE, although their gains are not additive or causal. Current BAN/BDNB and amenity snapshots are explicitly classified as retrospective reconstruction; only dated DPE and lagged comparable-sale variables satisfy full point-in-time rules. The result shows a meaningful but bounded improvement and does not turn the AVM into a certified appraisal.

Keywords: automated valuation model; DVF; BAN; BDNB; DPE; entity resolution; point-in-time features; CatBoost; Paris real estate.

1 Executive result

Phase 3 was designed to answer one practical question: does adding reliable property identity, building condition proxies, energy information, access and neighborhood context improve out-of-year valuation accuracy beyond Phase 2? The answer is yes on the fixed 2025 cohort. The selected model reduces MAE by EUR 3,293 (3.41%), improves MdAPE from 12.22% to 11.70%, increases the share of estimates within 20% from 71.69% to 73.39%, and raises R^2 from 0.8601 to 0.8679. The paired interval for the MAE reduction excludes zero.

The contribution is broader than one score. It includes a versioned Bronze source layer, deterministic address and building resolution, a row-level match audit, a leakage-aware DPE join, an enriched 141-column Gold table, controlled feature-family ablations, a reusable CatBoost artifact and an auditable single-property inference path.

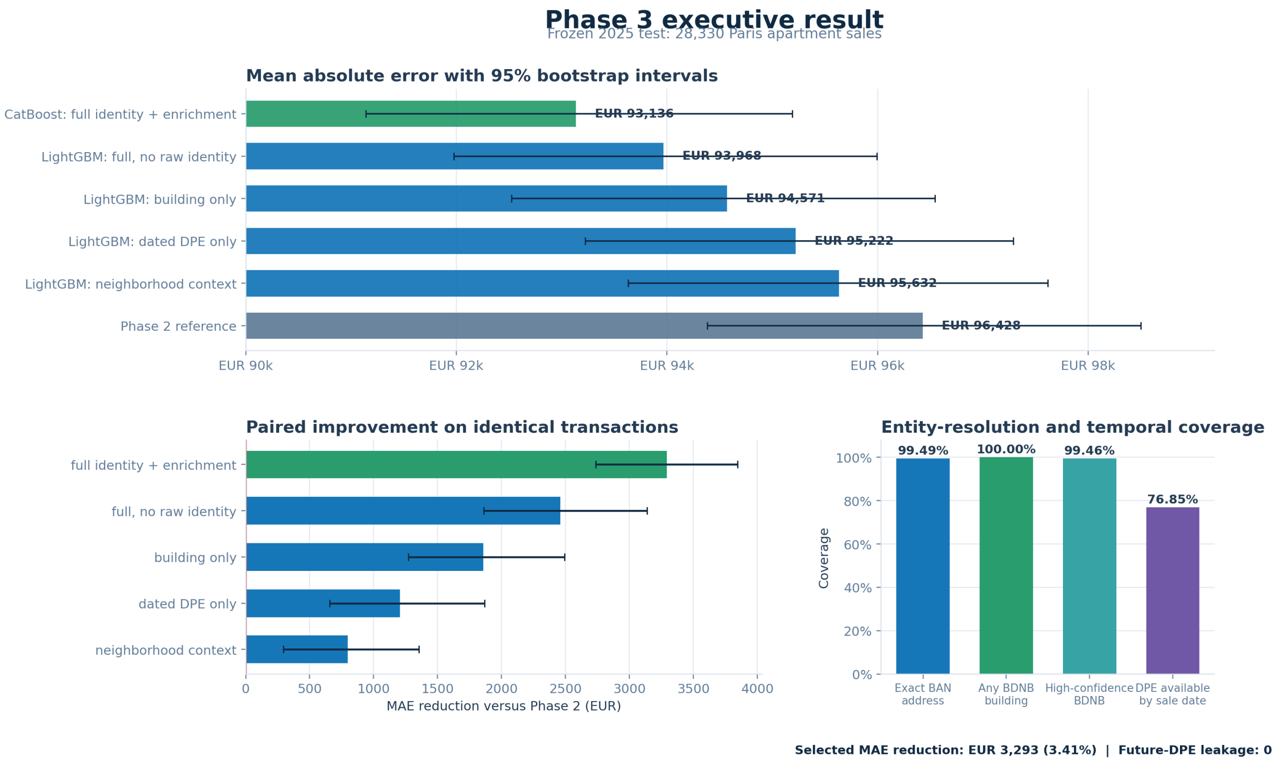

Figure 1: Executive Phase 3 result. Error bars are nonparametric 95% bootstrap intervals. Paired improvement intervals resample transaction-level differences against the same frozen Phase 2 predictions.

2 Problem setting and prior phases

The target is total transaction value in euros for an ordinary apartment sale in Paris. Phase 1 established a chronological tabular-model benchmark using DVF characteristics. Phase 2 added Lambert-93 spatial variables, H3 cells and historical comparable-sale statistics under a minimum 90-day information lag. Its best method combined 75% of a LightGBM estimate with 25% of the trailing 365-day, 500-metre local median price per square metre. That correction reached EUR 96,428 MAE in 2025, improving the Phase 1 reference only modestly.

Phase 2 also exposed the main information gap. DVF describes the transaction and apartment size but not the building, dated energy state, exact persistent address identity, transit access or environmental context. Phase 3 therefore focuses on external data linkage rather than a more exotic neural architecture.

3 Data and external sources

3.1 Fixed transaction cohort

The transaction cohort is unchanged from the earlier phases. DVF rows are collapsed and filtered at mutation level so that the target is not duplicated across property lines [1]. The cohort retains arm's-length sales containing exactly one apartment, while allowing dependencies such as cellars or parking spaces. Required fields include date, value, surface and Paris coordinates. Fixed plausibility bounds are 9–300 m^2 for surface, EUR 50,000–10,000,000 for value and EUR 2,000–30,000 per m^2.

Table 1: Chronological cohorts used throughout Phase 3.

Period	Sales	Cumulative training sales	Role
2021	30,397	30,397	Development
2022	33,080	63,477	Development
2023	26,798	90,275	Development
2024	24,404	114,679	Configuration selection
2025	28,330	—	Frozen chronological test
Total	143,009	—	Fixed Gold identity

3.2 Authoritative enrichment sources

Table 2 summarizes the new inputs. Every downloaded artifact is stored under `data/bronze/` with producer, licence, source URL, retrieval timestamp, byte size and SHA-256 recorded in `reports/phase3/source_manifest.json`. The BDNB departmental archive is version 2026-02.a for Paris and contains both DPE records and the official DPE-to-building relation; a second bulk ADEME download is therefore not needed.

Table 2: Phase 3 external sources and snapshot interpretation.

Source / producer / licence	Variables introduced	Temporal status
BAN [2]		
IGN, Licence Ouverte 2.0	Normalized address ID, FANTOIR road identity, coordinates and cadastral references	Current static reference
BDNB 2026-02.a [3]		
CSTB, Licence Ouverte 2.0	Building ID, footprint, levels, height, year/period, dwelling and lot counts, materials and coarse state	Retrospective static reconstruction
DPE in BDNB [4]		
ADEME/CSTB, Licence Ouverte 2.0	Energy and GHG classes, consumption, emissions, heating, surface and diagnostic date	Point-in-time date filter
Rail stations [5]		
Île-de-France Mobilités, Licence Ouverte 2.0	Metro, RER, train and tram distance/density by mode and line	Current static context
Schools and parks [6]		
Ville de Paris, ODbL	Nearest distance, counts and green-space area	Current static context
Services		
BDNB BPE, Licence Ouverte 2.0	Nearby shop/service distance and counts	Retrospective static context
Noise [7]		
DRIEAT/CEREMA, Licence Ouverte 2.0	2022 road, RATP-rail and SNCF-rail Lden bands	Fixed 2022 strategic map

4 Data architecture and entity resolution

The pipeline follows Bronze–Silver–Gold layering (Figure 2). Bronze preserves source snapshots. Silver resolves identities and records how every match was obtained. Gold retains one row per original DVF sale and adds the external fields plus explicit missingness and provenance indicators.

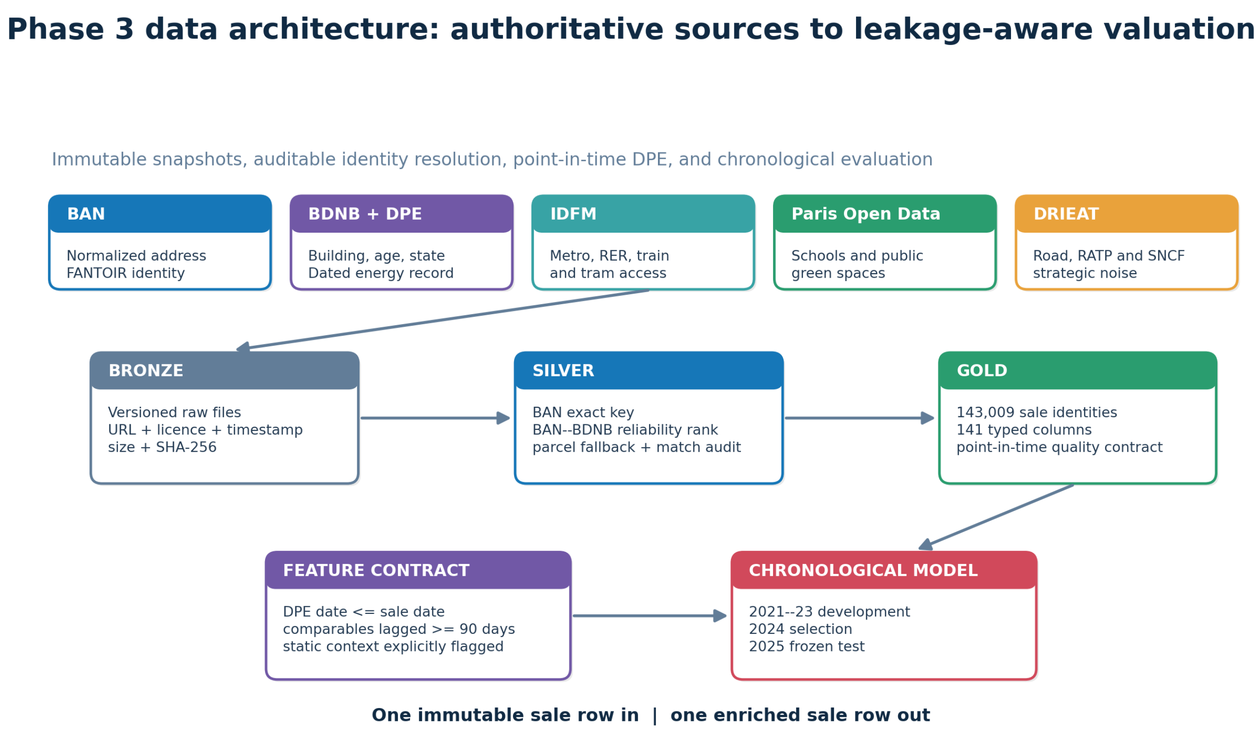

Figure 2: Source-to-model architecture. The feature contract distinguishes dated information from retrospective static reconstruction.

4.1 Exact address identity

The primary address key combines commune, the four-character FANTOIR/DVF road code, street number and normalized suffix. It is deterministic and avoids fuzzy street-name matching. The key resolves to the BAN interoperability ID. The resulting exact BAN match rate is 99.49%. Coordinates and labels from BAN are retained for audit, not used to overwrite the original transaction identity silently.

4.2 BAN-to-BDNB building resolution

The official BAN–BDNB relationship table can return several buildings for one address. Candidates are ranked using the relationship reliability field; the highest-ranked building is selected deterministically. When the address route is unavailable, the cadastral parcel relation is used as a fallback. The Gold table records the selected BDNB building, match method, confidence, number of candidates and both address and parcel evidence. This produces a 99.997% building match rate and 99.46% high-confidence rate. The row-level Silver audit is stored in `data/silver/phase3_match_audit.parquet`.

4.3 Identity-preserving joins

The build asserts that the enriched table has exactly 143,009 rows, that the original mutation order and IDs are preserved and that no duplicate mutation is created. This matters because one-to-many building, DPE or amenity joins can otherwise inflate the training sample and create false accuracy. The delivered Gold table has 141 typed columns and a recorded SHA-256 contract.

5 Feature engineering and temporal contract

5.1 Building structure, age and condition

BDNB contributes construction year or period, age at sale, level count, height, footprint, dwelling count, total/parking lot counts, wall and roof materials, usage class and coarse building state. Construction age is computed relative to each sale year and invalid negative or implausible values are rejected. The median building age in the 2025 test is 125 years, reflecting the old Paris housing stock. These fields are valuable retrospective descriptors, but the 2026 snapshot does not prove that every value was observable in earlier years.

5.2 Point-in-time DPE

For a sale at date t, a diagnostic is eligible only when

$$\text{date\_etablissement\_dpe} \leq t. \tag{1}$$

Within each resolved building, an as-of join selects the latest eligible DPE. The pipeline records DPE age in days, apartment-surface discrepancy and match confidence. A hard validation rejects any future diagnostic. The observed future-DPE leakage count is zero.

Overall historical coverage is 76.85%, but it is strongly time-dependent: 13.12% in 2021, 83.38% in 2022, 97.10% in 2023, 99.05% in 2024 and 99.33% in 2025. This pattern is expected because the current DPE regime begins in the study period; it also explains why a single aggregate coverage percentage is insufficient.

5.3 Accessibility, amenities and noise

Coordinates are transformed to Lambert-93 before metric distance calculations. The new context variables include nearest Metro, RER, train, tram, school, green-space centroid and shop/service distances; counts of stations, modes, lines, schools, green spaces and services inside fixed radii; green-space area; and maximum intersecting road/RATP/SNCF Lden bands. In the 2025 cohort, median straight-line distances are 222 m to Metro, 946 m to RER, 154 m to a school, 103 m to a green-space centroid and 6 m to a BDNB BPE service location.

Noise values represent strategic contour bands rather than parcel-level acoustic measurements. Locations outside a mapped high-noise contour are represented by the low/no-band category and a missing-contour indicator. They must not be read as an exact measured decibel value.

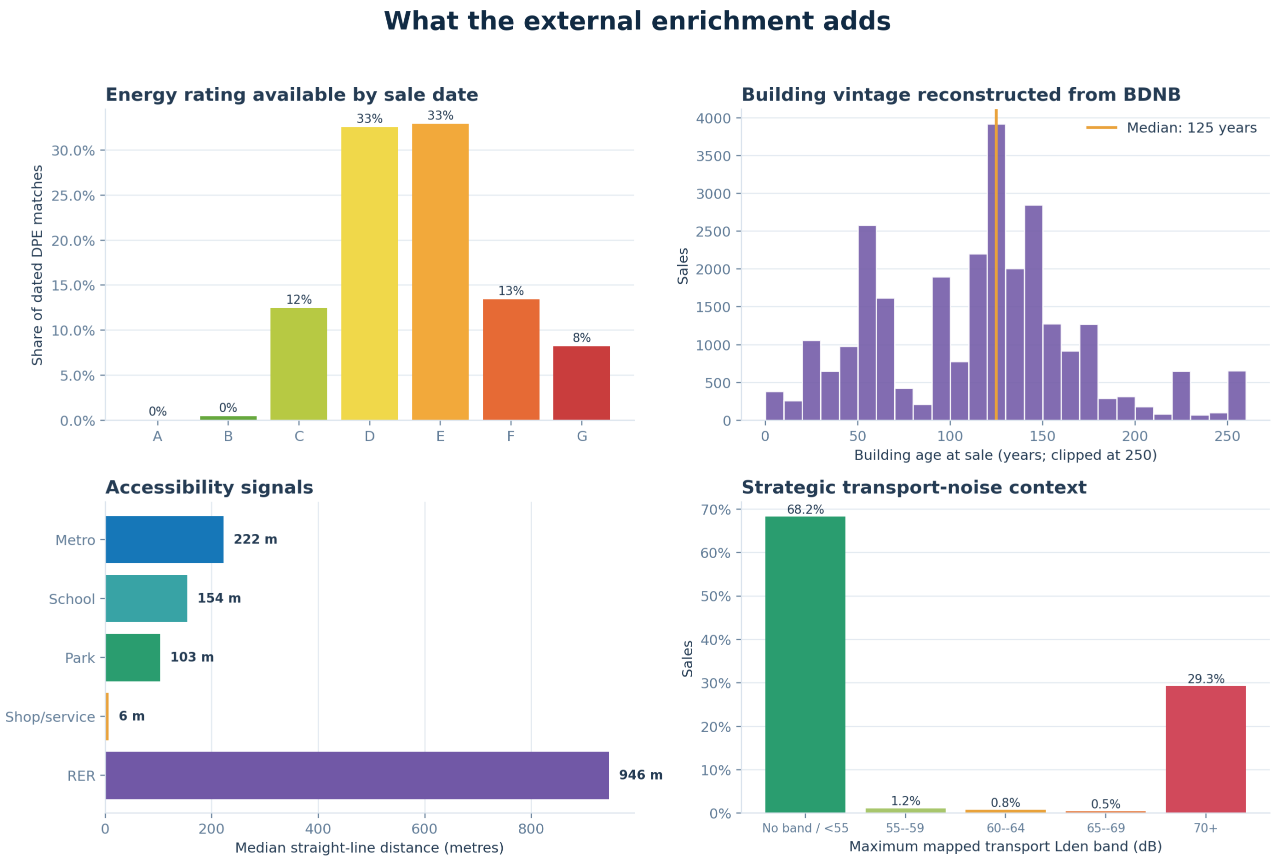

Figure 3: External-feature profile on the 2025 test cohort. DPE shares are conditional on a diagnostic available by the sale date. Distances are straight-line Lambert-93 measures. Noise is a strategic contour category.

5.4 Unavailable requested fields

The reviewed public BAN, BDNB and DPE fields do not provide a reliable apartment-floor or elevator variable for this cohort. Building level count is not a substitute for apartment floor. Gold therefore includes explicit `apartment_floor_available=0` and `elevator_source_available=0` flags while leaving the values missing. This avoids converting an unsupported assumption into model input. A lawful notarial, copropriété, property-manager or licensed-listing source would be needed to complete those fields.

5.5 Information-time classification

The final contract separates two evidence classes:

- **Point-in-time:** DPE dated on or before the sale and all Phase 2 comparable-sale features subject to a minimum 90-day publication lag.
- **Retrospective static:** current BAN identity, most BDNB physical attributes, current transport/school/park/service snapshots and the fixed 2022 strategic noise map.

The full model is consequently a strong retrospective reconstruction, not a claim that every external value was operationally available on each historical valuation date.

6 Modeling and evaluation

6.1 Feature-family ablations

All candidates retain the core DVF and Phase 2 variables so that Phase 3 tests incremental information. Five LightGBM feature sets are compared: Phase 2 only, dated DPE, building/linkage, neighborhood context and the full enrichment without raw high-cardinality identity. Each is evaluated with MAE and Huber objectives using a small predefined configuration grid. CatBoost additionally receives raw BAN address and BDNB building categories because its ordered categorical handling is suitable for high-cardinality identity. The final CatBoost configuration has 650 iterations, learning rate 0.05, depth 7, ℓ_2 leaf regularization 5 and MAE loss.

The CatBoost comparison changes both model family and access to raw identity. Its advantage over full LightGBM therefore cannot be attributed to identity alone. Likewise, feature-family ablation gains overlap and must not be added.

6.2 Chronological protocol

Development uses 2021–2023. The 2024 cohort selects the feature set, objective and hyperparameters. Selected candidates are refitted once on 2021–2024 and scored on the common 2025 rows. No 2025 record is used by a fitting or selection call. However, the 2025 cohort had already been examined in Phases 1 and 2; it is frozen for Phase 3 computation but not a pristine analyst-blind external test. A later prospective cohort is required for final deployment evidence.

6.3 Metrics and uncertainty

The primary metric is

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|. \tag{2}$$

Secondary metrics are RMSE, R^2, mean and median absolute percentage error (MAPE and MdAPE), and the shares within 10% and 20% of the transaction value. MAE intervals use nonparametric bootstrap resampling. Improvement intervals use paired resampling of per-sale absolute-error differences, preserving the fact that every candidate predicts the same transactions.

7 Results

7.1 Ablation and final test performance

Table 3 reports the principal 2025 results. Every MAE-objective enrichment ablation improves on the frozen Phase 2 reference. Dated DPE alone reduces MAE by EUR 1,206; building/linkage by EUR 1,858; neighborhood context by EUR 796; and the full no-identity LightGBM model by EUR 2,460. Their paired 95% intervals exclude zero. The selected CatBoost model improves MAE by EUR 3,293.

Table 3: Frozen 2025 chronological test. Intervals are paired MAE reductions against Phase 2; positive values favor the row method.

Method	MAE (EUR)	MdAPE	Within 20%	Reduction (95% CI), EUR
CatBoost, full identity/enrichment	**93,136**	**11.70%**	**73.39%**	**3,293 (2,735–3,846)**
LightGBM, full without raw identity	93,968	11.80%	73.03%	2,460 (1,860–3,137)
LightGBM, building/linkage	94,571	11.90%	72.28%	1,858 (1,271–2,493)
LightGBM, dated DPE	95,222	11.97%	72.28%	1,206 (657–1,867)
LightGBM, neighborhood context	95,632	11.99%	71.74%	796 (293–1,355)
Phase 2 comparable correction	96,428	12.22%	71.69%	Reference

For the winner, RMSE is EUR 192,806, $R^2 = 0.8679$, MAPE is 18.12%, 43.68% of estimates are within 10% and 73.39% are within 20%. Phase 3 reduces absolute error on 54.21% of individual test sales; the median transaction-level gain is EUR 1,496. The improvement therefore reflects many modest gains plus a smaller number of large corrections, not universal superiority on every sale.

7.2 Calibration, value segments and time stability

Figure 4 shows that the predicted and observed values remain tightly aligned over most of the range, with dispersion increasing among high-value sales. Phase 3 lowers MAE in every transaction-value decile shown, including the upper decile where absolute errors are much larger. Monthly 2025 MAE also remains below Phase 2 in every month. This supports temporal stability within the test year, although it does not measure a new market regime.

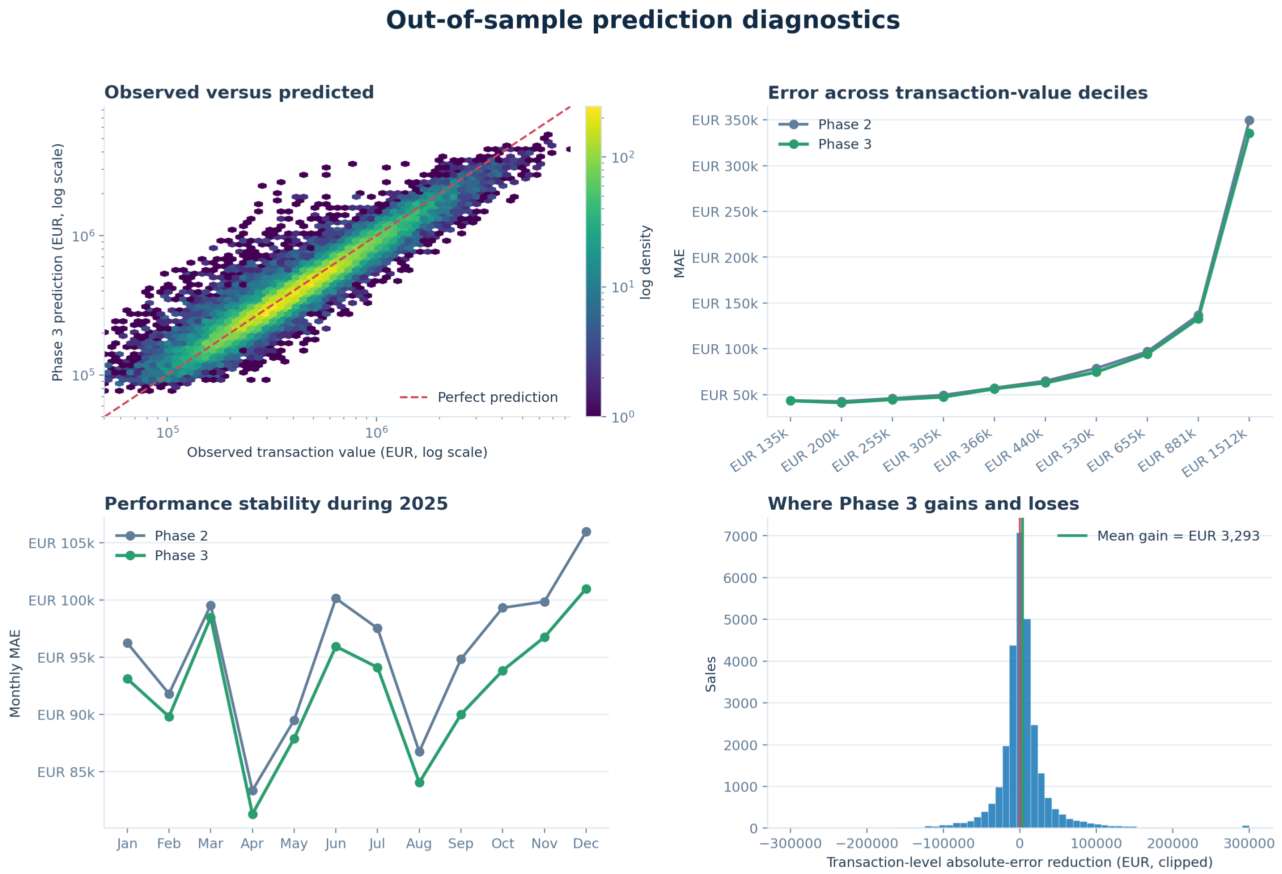

Figure 4: Frozen-test diagnostics. The transaction-level gain distribution is clipped for legibility; reported averages use the unclipped differences.

7.3 Geographic heterogeneity

The mean improvement is positive in 19 of 20 arrondissements. It is largest in the 6th, 4th, 7th, 8th and 1st arrondissements and approximately neutral in the 16th (EUR −167 mean change). The map also shows intermixed gains and losses within the same neighborhoods. This is consistent with heterogeneous errors at the individual property level and argues against interpreting arrondissement averages as causal amenity premiums.

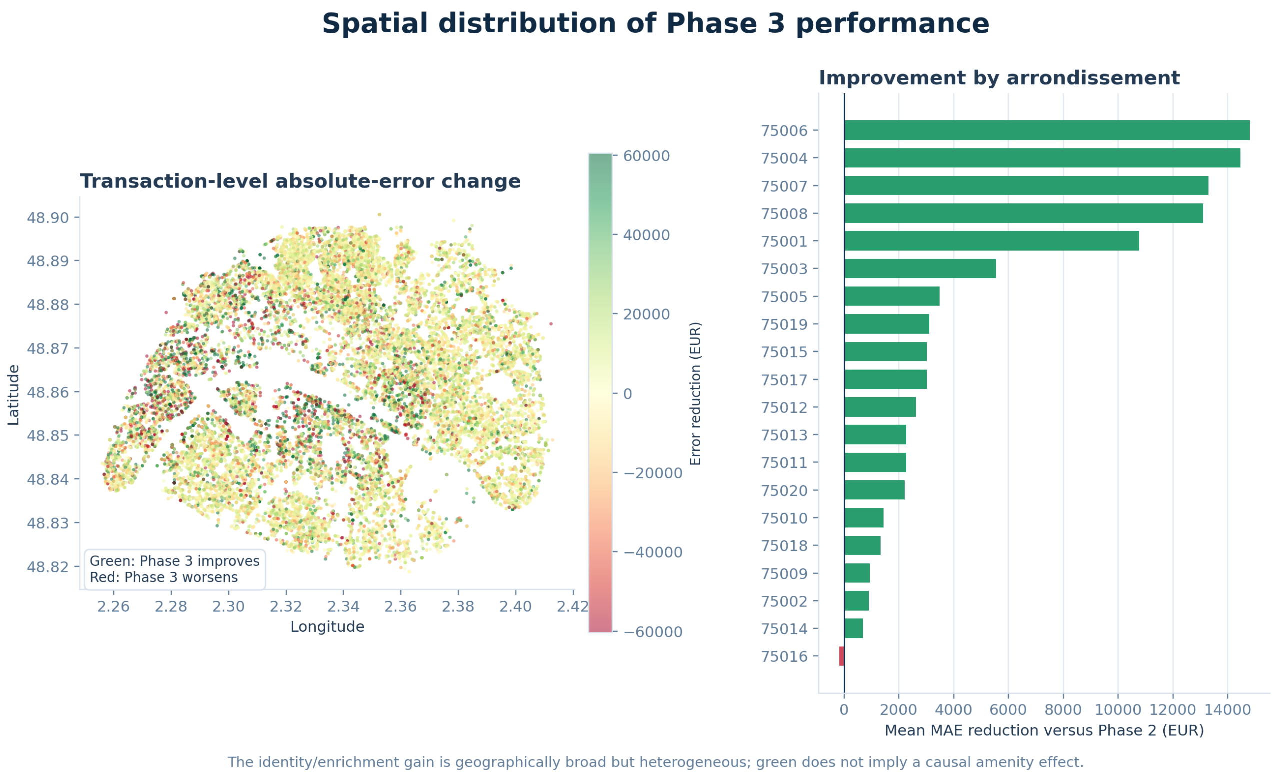

Figure 5: Spatial distribution of transaction-level changes and mean MAE reduction by arrondissement. Green marks lower absolute error under Phase 3.

7.4 Model use of enriched information

CatBoost prediction-value-change importance assigns 79.68% of total importance to core DVF/spatial variables, dominated by surface and log surface. Historical comparables account for 12.95%; building/linkage 2.30%; neighborhood context 2.01%; DPE/energy 1.86%; and identity/micro-location 1.20%. The strong ablation gains alongside small individual importances are not contradictory: external variables can refine difficult cases while surface still explains most price-level variation. Importance values describe the fitted model's use of features and do not estimate willingness to pay or causal effects.

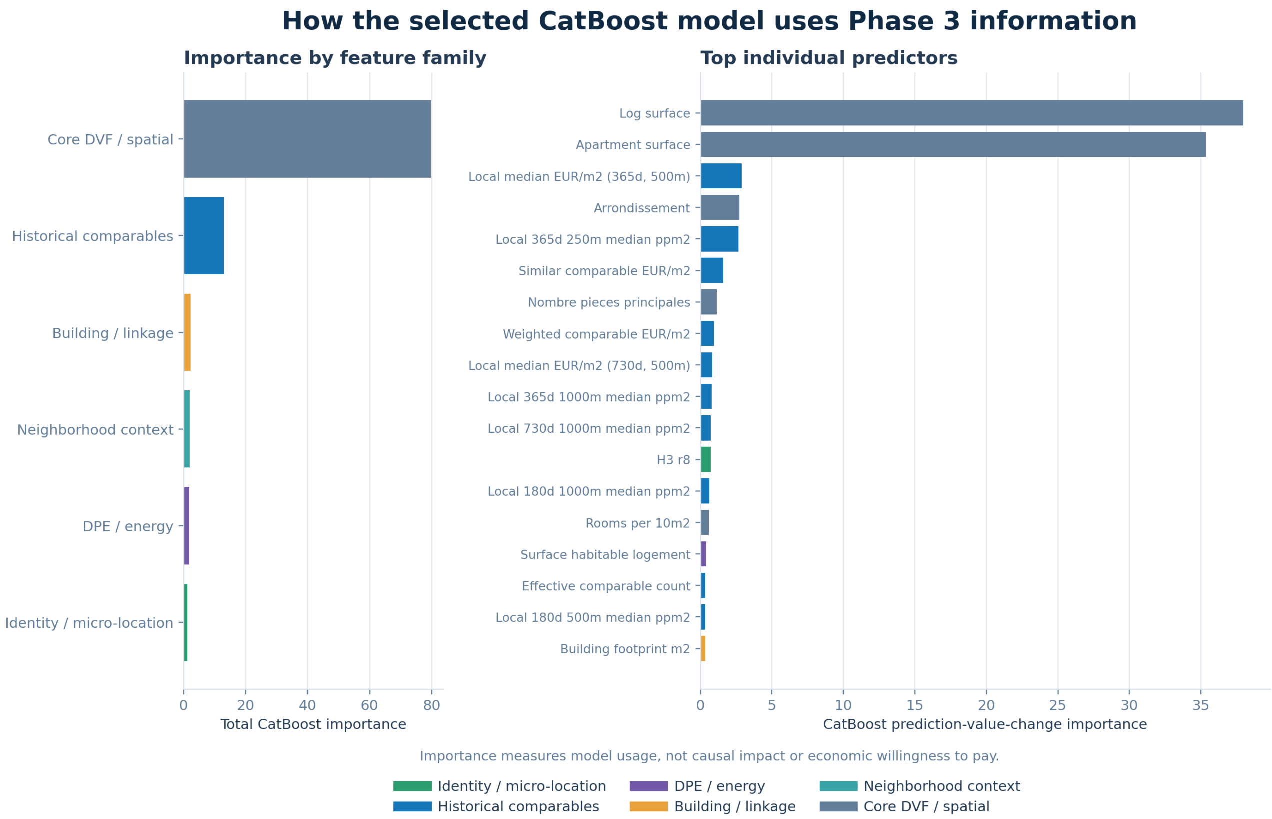

Figure 6: CatBoost prediction-value-change importance for the selected model. Feature-family totals sum to 100. Correlated variables may divide importance.

8 Interpretation and practical value

Three conclusions follow. First, external information addresses a real omitted- variable problem. Each enrichment family provides an independently positive test signal under the controlled LightGBM ablations. Second, reliable identity is operationally important even when its direct feature importance is small: it allows building, DPE and context records to attach to the correct transaction and enables CatBoost to learn repeated micro-location patterns. Third, the improvement is meaningful but bounded. A 3.41% MAE reduction does not remove unobserved apartment condition, renovation quality, floor, elevator, view, exposure, occupancy or private outdoor space.

The selected artifact supports single-property inference through `paris_avm.inference.phase3`. The caller supplies the apartment attributes, coordinates, valuation date, commune code and FANTOIR street code. The command resolves an exact existing BAN/BDNB identity, rolls DPE back to the valuation date, recomputes lagged comparable features and returns an automated indication. It also warns that building and neighborhood fields use the 2026 static snapshot.

9 Limitations and threats to validity

1. **Static-snapshot bias.** Current physical/context snapshots can improve reconstruction of past sales without proving historical availability. This is the most important limitation.

2. **Previously inspected test year.** The 2025 rows are excluded from Phase 3 fitting and selection, but earlier phases had exposed aggregate 2025 performance. Prospective confirmation is still necessary.

3. **Residual property heterogeneity.** Floor, elevator, interior condition, view and renovation quality remain unavailable.

4. **DPE linkage.** A building can contain several apartments and diagnostics. The latest eligible building-linked DPE may not be the exact apartment; surface discrepancy and confidence fields make this uncertainty visible but do not eliminate it.
5. **Noise and amenity semantics.** Straight-line distances ignore walking networks and barriers. Strategic noise contours are categorical planning layers, not indoor measurements.
6. **Bootstrap scope.** The paired interval measures sampling variation within the 2025 cohort, not uncertainty under future interest-rate, policy or market-regime shifts.
7. **Non-causal analysis.** Ablations and feature importance quantify predictive association only.

10 Reproducibility and transfer package

The complete Phase 3 transfer package is contained in the project:

- acquisition: `paris_avm.data.acquire_phase3`;
- entity resolution and features: `paris_avm.features.phase3`;
- ablation/model benchmark: `paris_avm.modeling.benchmark_phase3`;
- figures: `paris_avm.visualization.phase3`;
- inference: `paris_avm.inference.phase3`;
- Gold table: `data/gold/phase3_sale_features.parquet`;
- audit and contracts: `data/silver/phase3_match_audit.parquet` and `reports/phase3/`;
- selected model: `models/phase3/phase3_selected_model.joblib`.

The deterministic execution order is:

```
python -m paris_avm.data.acquire_phase3 --skip-dpe
python -m paris_avm.features.phase3
python -m paris_avm.modeling.benchmark_phase3
python -m paris_avm.visualization.phase3
powershell -ExecutionPolicy Bypass -File docs/paper/compile_phase3.ps1
```

The `-skip-dpe` option is intentional because the dated ADEME DPE rows and official DPE-to-building relation are already present in the versioned BDNB archive.

11 Conclusion

Phase 3 converts a transaction-only benchmark into an auditable building-aware valuation pipeline. Near-complete address/building resolution, strict dated-DPE selection and controlled feature-family experiments produce a statistically consistent EUR 3,293 MAE reduction on 28,330 sales from 2025. The strongest model remains driven by surface and historical market evidence, while external building, energy and context features provide useful refinements. The proper next step is not more retrospective tuning: it is prospective evaluation on a new cohort with vintage-correct snapshots, followed by lawful acquisition of apartment floor, elevator and condition data. Until then, the output should be used as a transparent automated indication rather than a certified appraisal.

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